

12(a)(i)	unstable nucleus	B1
	emits ionising radiation or decays spontaneously	B1
12(a)(ii)	probability of decay (of a nucleus)	M1
	per unit time	A1
12(b)	$A = \lambda N$	C1
	$560 = 9.9 \times 10^{-7} \times N$	A1
	$N = 5.7 \times 10^8$	
12(c)	$A = A_0 e^{-\lambda t}$	C1
	$170 = 560 \exp(-9.9 \times 10^{-7} \times t)$	
	$t = 1.2 \times 10^6 \text{ s}$	C1
	= 14 days	A1